Forecasting Rental Prices in Champaign

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Abstract
Accurately forecasting rental prices is crucial for financial planning and policymaking, especially in cities like Champaign, Illinois, where a large student population and academic-driven housing cycles shape the market. This study applies advanced time series analysis and machine learning methods to improve prediction accuracy, using a dataset that integrates economic, demographic, and real estate indicators. Linear regression effectively captured temporal patterns but struggled with robustness against anomalies. Tree-based methods, such as Random Forest and XG-Boost, failed to handle temporal dependencies and strong trends. Huber Regression addressed outliers more effectively, while Prophet, with its additive framework for trends and seasonality, achieved the best results. Prophet's Test MSE of 183.65 and Rolling MSE of 175.11 outperformed all other models. Its forecast for the first quarter of 2025 estimates Champaign's rental index between 1324.90 and 1330.11. This study demonstrates the value of combining robust statistical models with data-driven techniques to analyze dynamic rental markets. The findings offer practical insights for renters and policymakers and provide a replicable framework for similar cities.

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1 Introduction

The cost of living in Champaign presents a critical challenge for its residents, particularly the student community, where rental prices significantly influence financial planning and overall quality of life. For students, housing costs often consume the largest portion of their monthly budget, limiting their ability to allocate funds for education, food, and other essentials. Similarly, for local residents, escalating rental prices heighten economic pressures, potentially exacerbating social inequality and displacement. Accurate rental price forecasting is vital to mitigate these impacts, aiding students and residents in financial decision-making while providing policymakers with actionable insights to promote equitable and sustainable housing policies.

This study is centered around the following research question: How can time series analysis and modern data-driven methods enhance the accuracy of rental price forecasting in Cham-

paign to provide meaningful insights for individual and community-level decision-making? By focusing on Champaign's unique socioeconomic and demographic attributes, including its large student population and cyclical housing demand driven by academic calendars, this study aims to create a tailored and replicable framework for analyzing rental price trends in similar cities.

1.1 Context and Relevance

Recent advancements in forecasting methods, particularly in time series analysis and machine learning, have shown significant promise in predicting housing market trends. For example, robust estimation techniques, such as Huber regression, have proven effective in handling datasets with anomalies, which are common in housing price data[1]. Similarly, time series tools like Prophet have been demonstrated to capture complex seasonal and trend dynamics in real-world scenarios, offering new possibilities for analyzing cyclical markets such as Champaign's rental market[4]. Furthermore, ensemble methods like Random Forests have been shown to excel in capturing variable interactions in real estate price prediction[2].

Despite these advances, most existing studies have focused on larger metropolitan areas, leaving a gap in research for mid-sized cities like Champaign, where the housing market is heavily influenced by the academic cycle and the transient nature of the student population. This study addresses this gap by leveraging baseline forecasting methods alongside advanced tools like Huber regression and Prophet to assess their applicability in Champaign's rental market.

2 Exploratory Data Analysis

2.1 Data

Our study utilizes a comprehensive dataset incorporating variables from three major categories: economic indicators, demographic and social indicators, and real estate market indicators. These variables are selected based on their theoretical importance and empirical evidence demonstrating their relevance in housing market studies.

The response variable in our study is the Zillow Observed Rent Index (ZORI) for Champaign, which serves as a representative measure of rental price trends in the region. ZORI is a repeat-rent index, calculated by averaging listed rents within the 35th to 65th percentile range for all homes and apartments. This approach ensures the index is representative of the broader rental market rather than being skewed by outliers. By focusing on the "All homes" category, ZORI provides a comprehensive view of Champaign's rental market, making it ideal for capturing long-term trends and informing predictive models.

Economic indicators provide insights into the macroeconomic environment affecting rental markets. Consumer Price Index (CPI) measures inflation, directly impacting rental affordability. Average Hourly Earnings in Champaign reflect income dynamics, while the 10-Year Real Interest Rate influences the relative attractiveness of renting versus homeownership. Garner and Verbrugge (2009) highlighted how inflation and wages shape housing affordability and rental price trends, emphasizing the importance of including such variables in predictive models[3].

Demographic and social indicators capture demand-side factors. UIUC student numbers reflect cyclical housing demand influenced by the academic calendar, while the Champaign-Urbana resident population represents broader housing demand. Addition-

ally, the poverty rate highlights economic disparities affecting affordability. Smith and Munro (2008) demonstrated the significance of student population fluctuations in shaping rental markets, particularly in university towns like Champaign[6].

Real estate market indicators reveal supply-side dynamics. The residential vacancy rate indicates housing availability, while new private housing units track supply growth. Median days on market gauge demand intensity, with shorter durations suggesting upward pressure on rental prices. Rosen and Smith (1983) introduced the concept of the natural vacancy rate, underscoring its role in rental market adjustments[5].

In addition to these categories, we include COVID-19 new death data as an external impact variable to capture disruptions during the pandemic. This variable reflects how major external shocks influence housing market dynamics, particularly in unprecedented times.

2.2 Frequency Aggregation

The frequency aggregation process ensures that datasets with varying time frequencies are converted into a consistent monthly frequency for analysis. Yearly data is forward-filled to populate all months within a year, while semesterly data is resampled with forward-filling to extend values across all months in each semester. Daily data is aggregated by taking the mean of daily values within each month. Additionally, certain variables, such as unemployment and homeownership rates, are converted from percentages to decimals for consistency. Missing values are addressed using regression imputation with IterativeImputer, ensuring that relationships among variables are leveraged to estimate missing points. A specific handling method is applied to the "new_deaths" variable, where missing values at the start are treated as zeros. The data is finally truncated to the period from July 2016 to July 2024, providing a uniform and clean dataset ready for time series modeling. This process ensures that trends and seasonal patterns across variables are preserved while enabling robust modeling.

2.3 Motivating Figures

The seasonal decomposition plots for "Champaign" and "Urbana" reveal strong upward trends, regular seasonal patterns, and relatively small residuals, highlighting the importance of incorporating trend and seasonal dummy variables in regression models to better capture these systematic patterns. The ACF plot for "Champaign" exhibits a slow, gradual decay, indicating strong autocorrelation and the presence of persistence in the data. The PACF plot, however, shows a significant spike at lag 1, followed by relatively small values, suggesting that the lag-1 response captures most of the direct relationship. This pattern provides strong motivation for incorporating lagged response variables in the regression model to account for the temporal dependence and improve predictive accuracy.

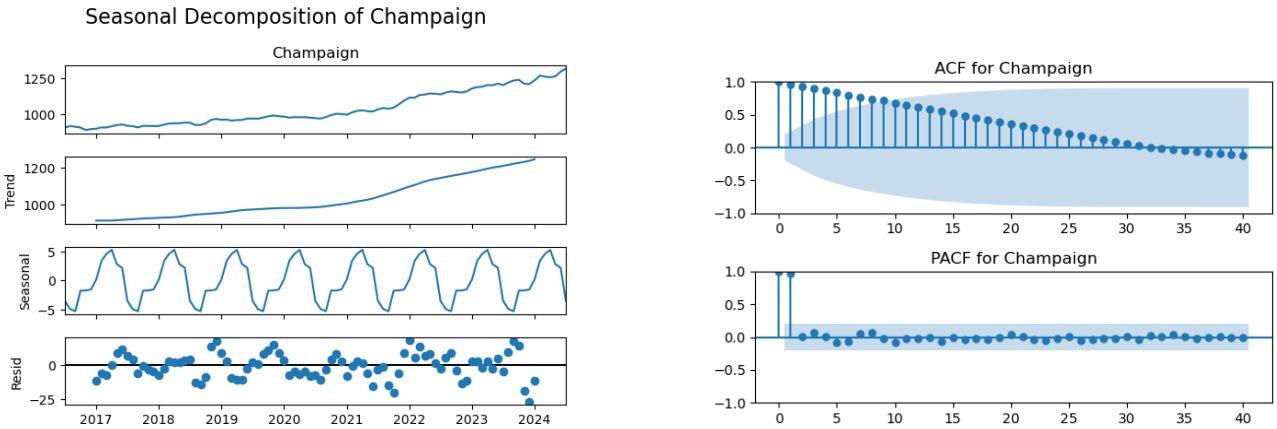

Figure 1: Seasonal decomposition (left) and ACF & PACF plots (right) for the response

The time series plots highlight the dynamics of various socioeconomic and housing-related variables from 2016 to 2024. Notably, many series exhibit anomalies during the COVID-19 pandemic (2020–2022), reflecting significant disruptions. For example, variables such as unemployment, new housing, and Illinois employees display pronounced spikes or drops during this period. Meanwhile, CPI and interest rates indicate shifts in economic policies and market conditions. The response variables, "Champaign" and "Urbana," show an overall upward trend, suggesting steady growth in housing prices or rental indices, with noticeable acceleration post-2021. These trends underscore the recovery phase and potentially long-term structural changes in housing and economic conditions. The distinct patterns in these time series offer critical insights into the relationship between macroeconomic factors and housing market dynamics. In the following analysis, only "Champaign" is considered as the main response variable.

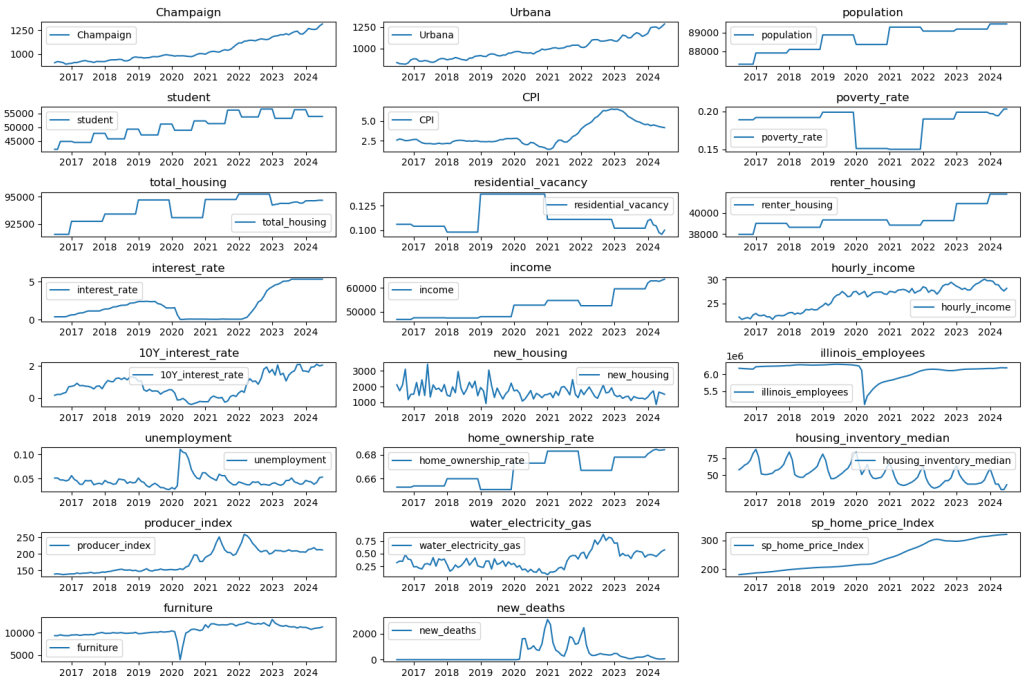

Figure 2: Time series plots of key variables from 2016 to 2024.

3 Baseline Models

3.1 Linear Regression

To model rental prices, linear regression was chosen as a baseline approach due to its simplicity and interpretability. The model incorporates a time trend and monthly dummy variables to capture seasonality. Additionally, lagged values of the response variable are included to account for temporal autocorrelation, as observed in the ACF and PACF plots.

Two specifications were tested: a linear trend model and a quadratic trend model. Statistical criteria such as Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Test Mean Squared Error (MSE) were used to evaluate model performance. The adjusted R^2 values for both models were exceptionally high, with the linear trend model achieving 0.988 and the quadratic trend model achieving 0.989. However, the quadratic model demonstrated signs of overfitting, with a significantly higher Test MSE (738.05) compared to the linear model (336.92). Consequently, the linear trend model was selected for its better generalization and lower complexity.

The chosen specification is as follows:

$$Y_t = \beta_0 + \beta_1 t + \sum_{m=1}^{11} \gamma_m D_m + \delta_1 Y_{t-1} + \sum_{k=1}^{21} \theta_k X_{k,t} + \epsilon_t$$

where Y_t is the ZORI value for Champaign at time t, serving as the response variable. t represents the linear time trend to capture long-term changes. D_m are monthly dummy variables ($m = 1, \ldots, 11$) used to account for seasonality effects. Y_{t-1} is the lagged response variable to account for temporal autocorrelation identified in the ACF and PACF plots. $X_{k,t}$ are additional independent variables ($k = 1, \ldots, 21$), capturing key economic, demographic, and market-specific factors influencing rental prices. ϵ_t is the residual term representing unexplained variations in rental prices.

3.2 Decision Tree and KNN

To compare against linear regression, tree-based models and K-Nearest Neighbors (KNN) regression were applied. Both models, however, struggled to handle the strong upward trend inherent in the data. To address this, first-order differencing was used to remove the trend and stabilize the series, allowing the models to predict changes rather than absolute values.

For KNN regression, hyperparameter tuning identified the optimal configuration as 3 neighbors, a Manhattan distance metric, and a distance-based weighting scheme. The n-step ahead forecast resulted in a Test MSE of 5080.38, reflecting the model's difficulty in capturing global trends effectively.

The Decision Tree model, optimized with a maximum depth of 5, achieved better performance than KNN, with an n-step ahead Test MSE of 527.04. However, Decision Trees lack the ability to extrapolate, making them unsuitable for data with strong trends. By partitioning the input space into discrete regions and predicting within those partitions, the model struggles to generalize beyond the observed training data. This limitation is particularly pronounced in our dataset, where the response variable exhibits a clear upward trajectory, leading to poor predictions for unseen values.

Tree-based ensemble methods like Random Forests and Gradient Boosting are likely to face similar challenges unless specifically adapted for time series data with strong trends. Their reliance on split-based structures makes extrapolating long-term trends

inherently difficult in this context.

Overall, the results highlighted the limitations of non-linear methods like KNN and Decision Trees in handling data with strong temporal dependencies and trends. These findings further emphasized the robustness of the linear regression approach in this context, particularly for capturing long-term dynamics.

3.3 Forecasting Methods

Two forecasting approaches were implemented to evaluate model performance under different scenarios: n-step ahead forecasting and rolling (expanding) window forecasting.

n-step Ahead Forecasting:

This approach involves using the entire training dataset to predict multiple future points within a fixed horizon. It is particularly suitable for evaluating overall predictive accuracy without the need for dynamic updates. The linear trend model achieved a Test MSE of 336.92, demonstrating strong generalization capabilities and robustness in capturing the underlying rental price trends.

Rolling Window Forecasting:

- The initial training dataset includes the first 80% of observations.
- At each step:
  - A new observation from the test set is added to the training set.
  - The model is retrained on the extended dataset.
  - A single prediction is made for the next observation.
- This iterative process continues until predictions are generated for all test observations.

The rolling window approach dynamically adapts to the latest trends in the data, achieving a Rolling MSE of 198.51 for the linear model, significantly outperforming the n-step ahead MSE of 336.92. The rolling forecasts closely tracked the true values, capturing both the general upward trend and short-term variations effectively.

4 Machine Learning Models

4.1 Tree-based Methods

Tree-based ensemble models, such as Random Forest, XGBoost, LightGBM, and CatBoost, are popular in machine learning due to their ability to capture complex non-linear relationships and reduce overfitting through ensemble averaging or boosting mechanisms. In this study, Random Forest was selected as the primary representative of this class of models. Key hyperparameters, including the number of trees (`n_estimators`), maximum tree depth (`max_depth`), and minimum sample requirements (`min_samples_split` and `min_samples_leaf`), were optimized using GridSearchCV combined with a time series-aware cross-validation scheme.

Despite these optimizations, Random Forest struggled to produce accurate forecasts for our dataset. The n-step ahead Testing MSE was 8048.84, and the Rolling MSE was 737.79, reflecting the model's inability to capture the strong upward trend and temporal dependencies present in rental prices. This failure can be attributed to the fundamental design of tree-based models, which partition the input space into discrete regions and predict within these regions based on training data. Such an approach makes it difficult for these models to extrapolate beyond the observed range, especially in datasets like ours, where the target variable exhibits a continuous and pronounced upward trajectory.

Moreover, the presence of anomalies in our data further exacerbated the limitations of tree-based models. Outliers can disproportionately influence the splitting criteria in decision trees, leading to over-specialized partitions and reduced generalization. Ensemble methods such as Random Forest attempt to mitigate this by averaging multiple trees, but the underlying structural issues remain. These problems are not unique to Random Forest but extend to other tree-based ensemble methods like XGBoost, LightGBM, and CatBoost. All these models rely on similar split-based architectures, which inherently lack the sequential awareness needed to model temporal dependencies effectively. Consequently, they struggle with datasets characterized by strong trends and irregularities.

4.2 Huber Regression

Huber Regression was employed to address the significant outliers introduced during the COVID period in our dataset. This method combines the robustness of L1 loss with the efficiency of L2 loss, offering a balanced approach to handling outliers while maintaining precision for inlier data. By adapting to the scale of residuals, Huber Regression mitigates the impact of extreme errors, making it particularly suited for datasets with anomalies like ours.

The core of Huber Regression lies in its loss function, defined as:

$$L_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{if } |a| \leq \delta \\ \delta \cdot (|a| - \frac{1}{2}\delta) & \text{if } |a| > \delta \end{cases}$$

where $a = y - f(x)$ represents the residual, and δ is a threshold parameter. This piecewise function transitions between L2 loss (for small residuals) and L1 loss (for large residuals), ensuring robustness to outliers while preserving sensitivity for smaller deviations. The choice of δ plays a critical role in balancing this trade-off: a smaller δ increases resistance to outliers but may compromise precision for inliers, while a larger δ behaves more like L2 loss, enhancing inlier accuracy but leaving the model vulnerable to outliers.

In our study, $\delta = 1.5$ was determined as the optimal value through GridSearchCV, which systematically explored a range of thresholds using time series cross-validation. This value achieved the best balance between robustness and precision, minimizing both Test MSE and Test MAE across different forecasting methods.

The Huber Regression model demonstrated significant improvements in predictive accuracy compared to previous approaches. For the n-step ahead forecast, it achieved a Test MSE of 456.52 and a Test MAE of 16.63. The rolling forecast further highlighted its robustness and adaptability, with a Rolling MSE of 186.91 and a Rolling MAE of 11.46, marking the best performance among all tested models. These results underscore the model's ability to handle both the upward trend and anomalies present in the data, outperforming other approaches in terms of both generalization and short-term prediction accuracy.

The model's effectiveness can be visualized in Figure 3, where the n-step ahead and rolling forecasts closely align with the true values, capturing both the overall trend and short-term variations. The rolling forecast, in particular, showcases Huber Regression's capacity to dynamically adapt to recent data, making it highly effective for time series forecasting tasks with challenging characteristics.

Overall, Huber Regression's ability to balance robustness and precision, combined with its superior performance in capturing the dataset's upward trend and mitigating the impact of outliers, positions it as the second-most effective model in this study.

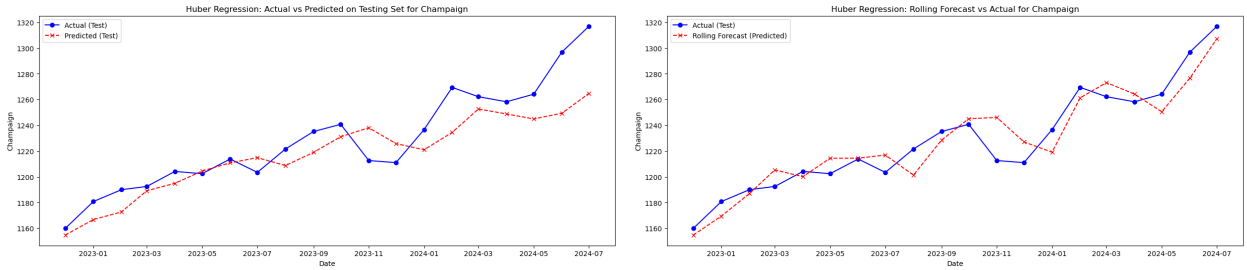

Figure 3: Left: n-step ahead forecast. Right: Rolling forecast.

4.3 Prophet

Prophet is a forecasting tool specifically designed for time series data characterized by strong trends and seasonal components. It models time series using an additive structure:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where $g(t)$ represents the trend component, $s(t)$ captures periodic seasonal patterns, $h(t)$ accounts for irregular but recurring holiday effects, and ϵ_t denotes the noise term.

The trend component $g(t)$ allows flexibility through the use of piecewise linear trends. These trends are segmented by changepoints, at which the growth rate can change. The mathematical representation of the trend is:

$$g(t) = (k + a(t)^T\delta)t + (m + a(t)^T\gamma)$$

Here, k is the baseline growth rate, m is the initial offset, and $a(t)$ is an indicator function that equals 1 if t is after a changepoint and 0 otherwise. The parameters δ and γ represent adjustments to the growth rate and offset at each changepoint. These parameters are regularized using a normal distribution:

$$\delta \sim \mathcal{N}(0, \sigma^2_{\text{changepoint}})$$

The hyperparameter `changepoint_prior_scale` controls the variance $\sigma^2_{\text{changepoint}}$, influencing the model's flexibility in fitting abrupt changes in the trend. A larger value allows the model to adapt to sharper changes, while a smaller value enforces smoother transitions.

The seasonal component $s(t)$ is modeled using a Fourier series, allowing the capture of complex periodic patterns. The formulation is:

$$s(t) = \sum_{n=1}^{N} \left[a_n \cos\left(\frac{2\pi nt}{P}\right) + b_n \sin\left(\frac{2\pi nt}{P}\right) \right]$$

where P is the period of the seasonal cycle (e.g., 365.25 for yearly seasonality), and a_n, b_n are the Fourier coefficients. These coefficients are also regularized using a normal distribution:

$$a_n, b_n \sim \mathcal{N}(0, \sigma^2_{\text{seasonality}})$$

The hyperparameter `seasonality_prior_scale` determines the variance $\sigma^2_{\text{seasonality}}$, controlling the model's ability to fit complex seasonal patterns. Larger values allow more intricate seasonal dynamics, while smaller values enforce simplicity.

The estimation of model parameters, including growth rate adjustments (δ), seasonal coefficients (a_n, b_n), and offsets (γ), is performed using Bayesian inference. The Maximum A Posteriori (MAP) approach is employed to identify parameter values that maximize the posterior probability given the observed data and prior distributions, balancing data fit and regularization imposed by priors.

In our study, the optimal hyperparameters for Prophet were determined as

`changepoint_prior_scale` = 0.001 and `seasonality_prior_scale` = 1.0. These values provided an appropriate balance between trend flexibility and seasonal complexity. The model achieved remarkable accuracy, with an n-step ahead forecast Test MSE of 183.65 and Test MAE of 10.68, as well as a Rolling MSE of 175.11 and Rolling MAE of 11.09. These results underscore Prophet's ability to capture both long-term trends and periodic patterns while handling anomalies effectively.

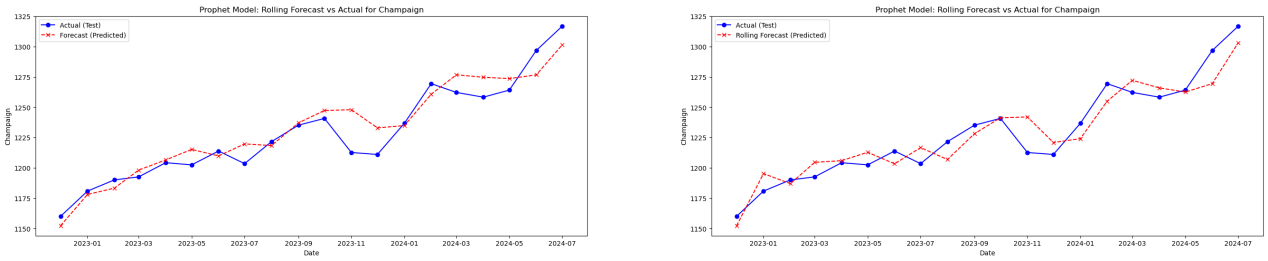

Figure 4: Left: n-step ahead forecast. Right: Rolling forecast.

5 Conclusions

This study investigated how time series analysis and modern data-driven methods can improve the accuracy of rental price forecasting in Champaign, a city with unique socioeconomic and demographic characteristics. Using various models and exploring their strengths and limitations, we developed a customized framework that addresses the city's cyclical housing demand, driven by its large student population and academic calendar.

Table 1 provides a comprehensive comparison of the forecasting models evaluated in this study. Among the baseline models, linear regression demonstrated robustness, achieving a Rolling MSE of 198.51. However, tree-based methods such as KNN and Decision Tree struggled with the strong upward trend and anomalies of the dataset, resulting in higher error metrics. Modern machine learning models, such as Random Forest and XG-Boost, faced similar challenges, primarily due to their inability to capture temporal dependencies and extrapolate trends effectively. Notably, Huber Regression and Elastic Net showcased improved performance, but Prophet emerged as the best-performing model across all metrics.

Prophet's ability to model strong trends and seasonal patterns, combined with its robustness to anomalies, proved critical to forecast rental prices in Champaign. Its rolling MSE of 175.11 and Rolling MAE of 11.09 were the lowest among all models, confirming its suitability for this context. Using the Prophet model, we forecast the Zillow Observed Rent Index (ZORI) for Champaign in early 2025 (January–March) to range between 1324.90 and 1330.11. This projection aligns with the observed upward trend in rental prices and reflects the continued influence of the city's socioeconomic dynamics.

This study successfully addressed the research question by demonstrating how advanced time series methods can enhance rental price forecasting in Champaign. Prophet, in particular, offered a tailored and replicable framework for capturing the city's cyclical housing demand and demographic-specific factors, such as its large student population. The methodological

insights and forecast accuracy achieved here can guide individual budget decisions and inform policy makers in cities with similar characteristics.

Table 1: Performance Comparison Across Models

Model	**Test MSE**	**Test MAE**	**Rolling MSE**	**Rolling MAE**
Baseline Models				
Linear Regression	336.92	13.15	198.51	11.61
KNN*	5080.38	63.96	268.89	15.10
Decision Tree*	527.04	15.94	353.54	15.00
ML Models				
Elastic Net	289.15	13.28	229.55	11.50
Random Forest	8048.84	78.92	737.79	20.05
XG-Boost	12991.74	96.88	538.92	18.45
LightGBM	14067.64	110.64	748.87	22.17
Huber Regression	456.52	16.63	186.91	11.46
Prophet	183.65	10.68	175.11	11.09
DL Models				
MLP (Univariate)	527.46	–	300.96	–
MLP (Multivariate)	3644.60	–	–	–
LSTM (Univariate)	281.38	–	–	–

Note: Models marked with * used the first differenced response variable.

In conclusion, the results highlight the value of combining robust statistical models, careful hyperparameter tuning, and domain-specific considerations to address complex forecasting problems. Future work could expand on these findings by integrating external regressors or exploring hybrid and deep learning methods to further improve predictive performance.

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